

Voight-Kampff 2.0

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ABSTRACT

"Voight-Kampff 2.0" is a reflection on the sustainable development of artificial intelligence technology in the future. Once a machine becomes conscious, would AI stay as a tool or obtain a soul? Would there be a boundary between human and machines?

A Voight-Kampff test process involving two individuals is simulated in this work, where the emotional fluctuations generated by empathy provide the basis for judging whether the suspect is human or machine. There is a saying in Buddhist philosophy that "what can think is the heart", which means that only equipped with empathy could one be considered with having personality and soul. Unlike breathing or heartbeat, EEG data is the most primitive and cannot be subjectively controlled, which also metaphors the process of "thinking". Therefore, the work chooses EEG data to test whether empathy produces emotional fluctuations.

"Voight-Kampff 2.0" aims to create emotionally charged experiences through biological data. With the changing images visualization of brain waves, the human facial image is artistically processed in a retro pixel style. Specifically, when a question is delivered by the judge, the emotional changes of suspect is stimulated and reflected in EEG data, which is visualized clearly in graphics and facial images on screen of both sides. The Match degree between emotion reflection and correct answer will help the judge determine whether the suspect is human or AI. Real human brainwave art images have obvious area changes and sound feedback due to the question, but machines don't. The work not only allows audiences to empathize with the work, but also founds a connection among audiences' emotion.

The work provides references for the creation project of EEG biological cooperation. Furthermore, significance of maintaining self-awareness in relationship of human-machine is also emphasized, which helps derive the sustainable development between human and AI.



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